Digital Logic Lab Project

Guessing the Word

Students:

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Introduction:

This project is a guessing word game, the user will use switches to play it, he will start the game, led 1 will turn on, then he will use the new-word switch so the word register will choose a random word of three letters, then he will set four switches which indicate the four bits of the letter, finally, he will enter the bits by setting enter switch to 1, and he has five trials, then the game logic circuit will compare if the entered letter is similar to one of the three letters of that in word register, if yes; the letter will be shown on a seven segment display, if not; no change will be done on the seven segment display but he will lose a trial, if the user guessed the three letters; Led 2 will be turned on, if he lost; Led 3 will be turned on.

System design:

Inputs:

- Start switch.
- New-word switch.
- Enter switch.
- 4 switches which indicate the 4-bits of the entered letter.

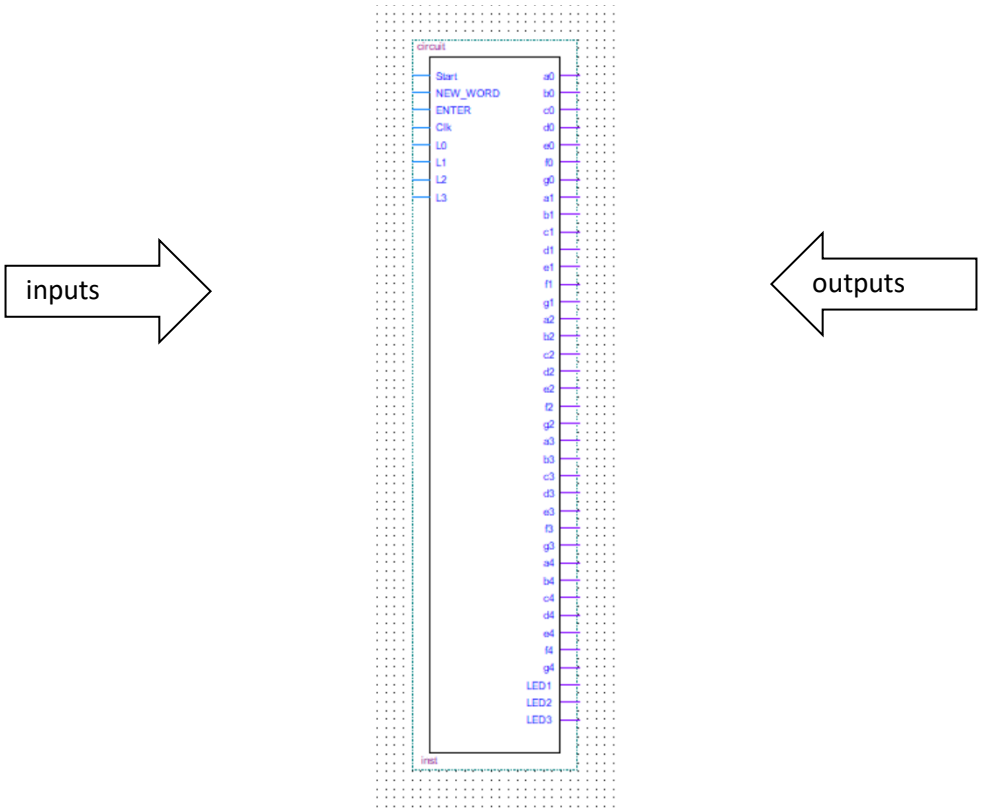
Outputs:

- Led 1 which will turn on when the user sets the start switch to 1.
- Led 2 which will turn on when the user wins by guessing the three letters correctly in less than 5 trials.
- Led 3 which will turn on when the user loses by running out of his 5 trials without guessing the word.
- Letter 7-segment display which will show the letter entered by the user.
- Guess 7-segment display which will show the correct letters entered by the user and matched the random chosen letters.
- Trials 7-segment display which will show the number of trials that are left for the user before he loses.

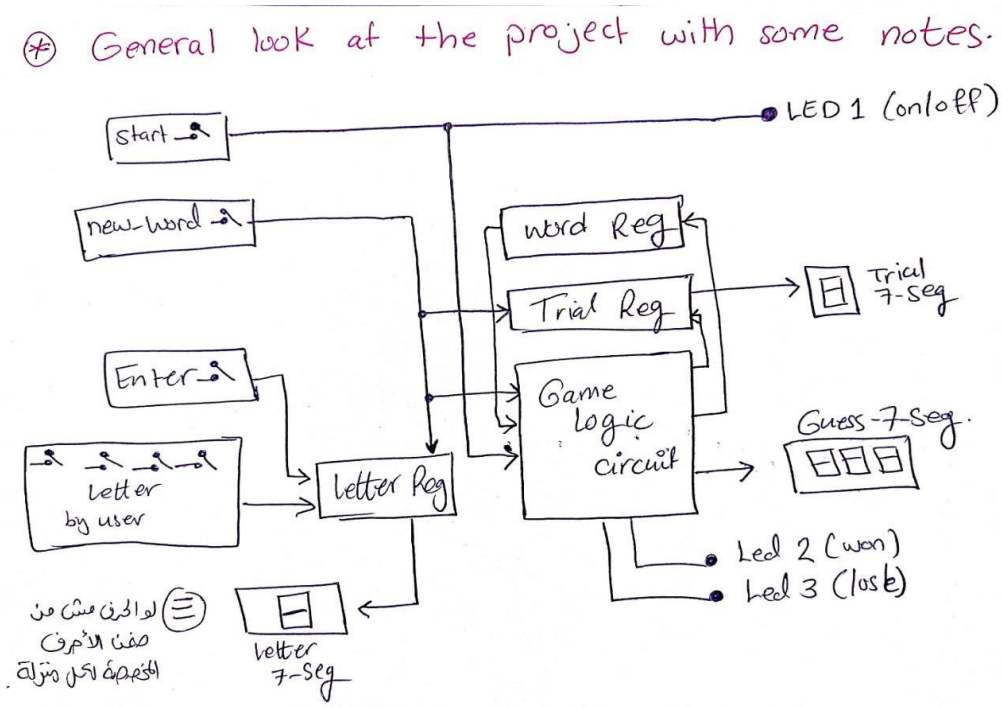
The system consists of the following subsystems:

- **Word register circuit**: a block of three counters, three registers, and three decoders, one for each letter (the random chosen ones), each register containing four D-flip-flops, each one will represent a bit of the letter. When the user set the new-word switch to 1, a random movement between letters will be done, then when he set the switch back to 0, the letters will be latched in the register.
- **Trial register**: has three D-flip-flops, the clock signal of this register is the comparator's output and reset signal is new-word switch.
- **Comparator**: contains 4 XNOR gates and one 4-bit AND gate to compare each bit of the entered ones by the user (using switches) with the random chosen ones if it's the same it will be shown on the register.
- **Letter register**: takes four inputs from the user using switches (which indicates the four bits of the letter) and controlled by enter switch.

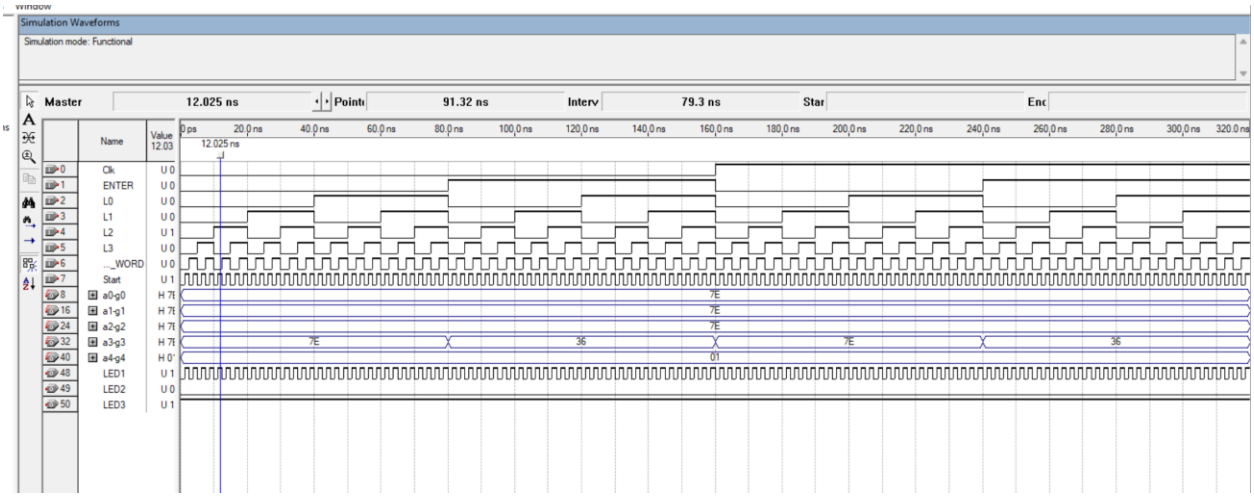
The full circuit in hardware system:



The state diagram:



Full circuit simulation:



System Testing and Results:

Word register:

Verilog code:

```
module wordRegCkt(Reset,clk,Q0,Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9,Q10,Q11,newword);
input Reset,clk;
output Q0,Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9,Q10,Q11,newword;

//both modules have same reset=new_word=1 and register's clock are trigered by new_word=0 and
//and so is signal newword of counter;

//module wordReg1(Reset,clk,C0,C1,C2,C3);
wordReg1(Reset,clk,C0,C1,C2,C3);
//module wordReg2(Reset,clk,C0,C1,C2,C3);
wordReg2(Reset,clk,C4,C5,C6,C7);
//module wordReg3(Reset,clk,C0,C1,C2,C3);
wordReg3(Reset,clk,C8,C9,C10,C11);

//module wordLet1Reg(C0,C1,C2,C3,Reset,clk,Q0,Q1,Q2,Q3);
wordLet1Reg(C0,C1,C2,C3,Reset,newword,Q0,Q1,Q2,Q3);
//module wordLet2Reg(C0,C1,C2,C3,Reset,clk,Q0,Q1,Q2,Q3);
wordLet2Reg(C4,C5,C6,C7,Reset,newword,Q4,Q5,Q6,Q7);
//module wordLet3Reg(C0,C1,C2,C3,Reset,clk,Q0,Q1,Q2,Q3);
wordLet3Reg(C8,C9,C10,C11,Reset,newword,Q8,Q9,Q10,Q11);

endmodule
```

Letter register:

Verilog code:

```
module  Letterckt(sw0,sw1,sw2,sw3,Reset,clk,a,b,c,d,e,f,g);
input sw0,sw1,sw2,sw3,Reset,clk;
output a,b,c,d,e,f,g;
//clk=enter;
//EN=enable in truth table=enter;

//module LetterReg(sw0,sw1,sw2,sw3,Reset,clk,Q0,Q1,Q2,Q3);
LetterReg(sw0,sw1,sw2,sw3,Reset,clk,Q0,Q1,Q2,Q3);

//module letter_segdriver(A,B,C,D,EN,a,b,c,d,e,f,g);
letter_segdriver(Q0,Q1,Q2,Q3,clk,a,b,c,d,e,f,g);

endmodule

```

Trial register:

Verilog code:

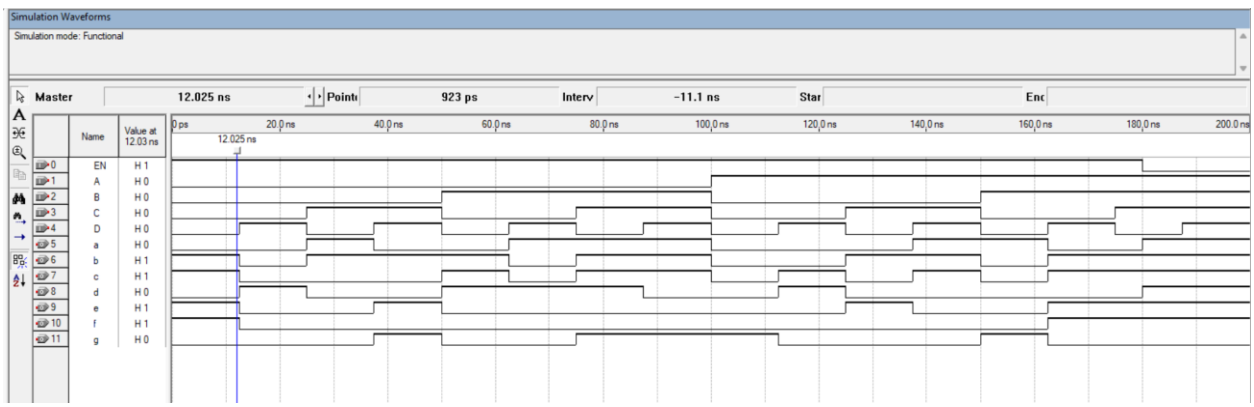
```
module trialsReg(P0,P1,P2,Reset,clk,Q0,Q1,Q2);
input P0,P1,P2,clk,Reset;
output Q0,Q1,Q2;
//clk is comparotor's output & reset is new word signal

//module dff1(D,Reset,clk,Q,Qnot);
dff1(P0,Reset,clk,Q0,Qnot0);
dff1(P1,Reset,clk,Q1,Qnot1);
dff1(P2,Reset,clk,Q2,Qnot2);


endmodule
```

Letter 7-segment:

Verilog code:

```
module letter_segdriver(A,B,C,D,EN,a,b,c,d,e,f,g);
input A,B,C,D,EN;
output a,b,c,d,e,f,g;
// ENABLE IS EXCLUDED HERE = ENTER SWITCH

assign a = (((~A&C&~D) | (~A&B&D) | (A&~B&C&D) | (A&B&~C&~D)) &EN )| ~EN;
assign b = (( C | (~A&~D) | (A&B&D)) &EN)| ~EN;
assign c = (((B&C) | (A&D) | (~A&~C&~D)) &EN)| ~EN;
assign d = (((~A&~C&D) | (~B&~C&D) | (~A&B&~D)) &EN)| ~EN;
assign e = (((A&C&~D) | (A&B&D) | (~A&~B&~C&~D)| (~A&~B&C&D)) &EN)| ~EN;
assign f=  (((A&B&D) |(A&B&C) | (~A&~B&~C&~D)) &EN)| ~EN;
assign g = ((~A&C&D) | (~A&B&C) | (A&~C&~D))& EN;

endmodule
```

Simulation:

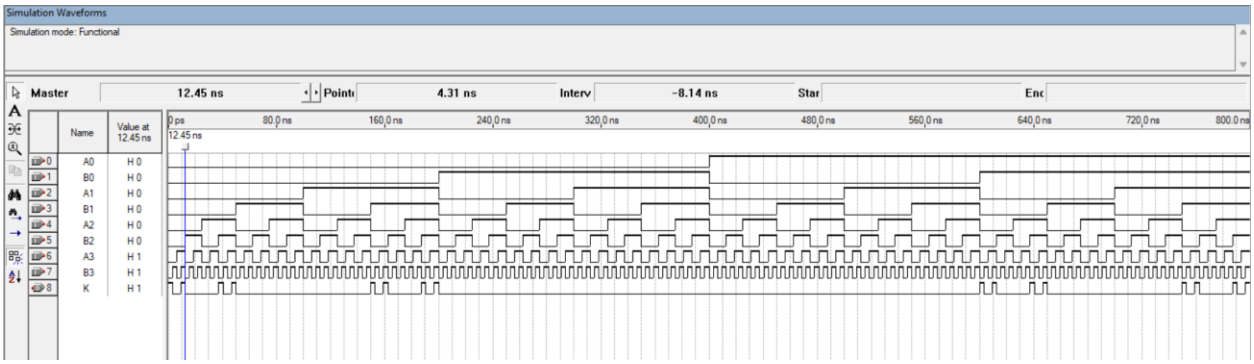


Comparator:

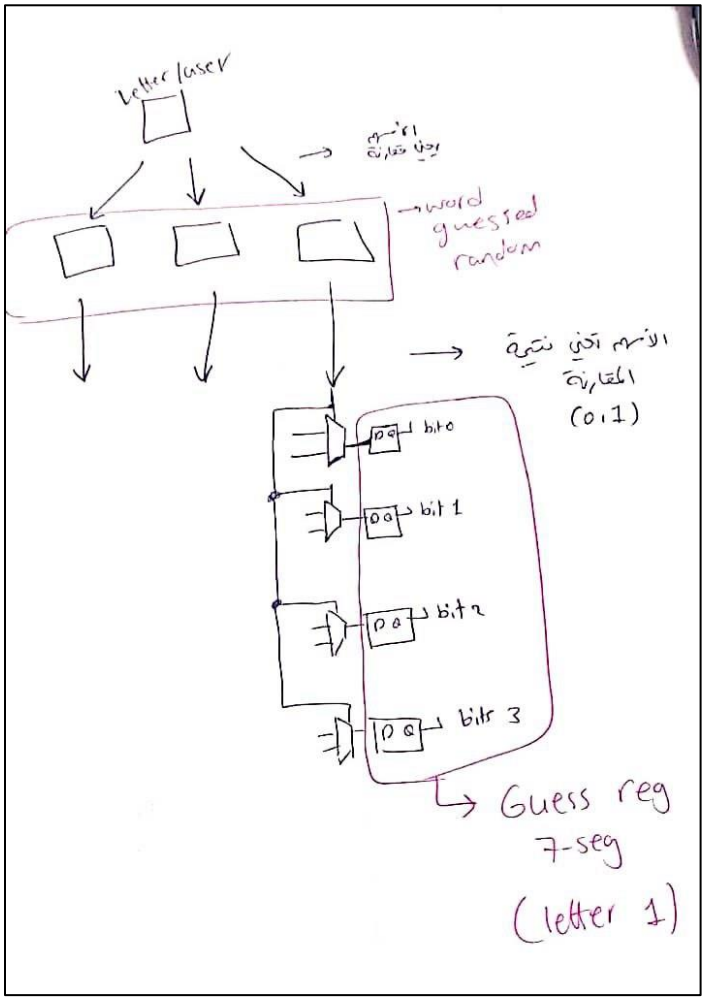
Verilog code:

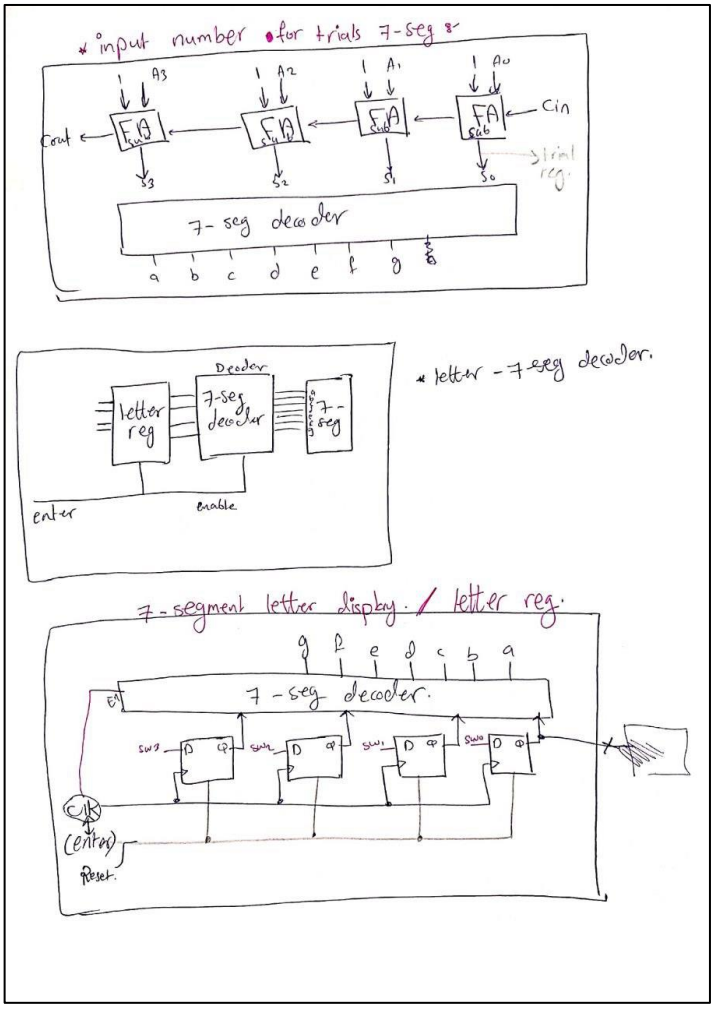
```
module comparator(A0,A1,A2,A3,B0,B1,B2,B3,K);
input A0,A1,A2,A3,B0,B1,B2,B3;
output K;
//COMPARES BITS IN ONE LETTER BETWEEN THE WORD GENERATED AND WORD ENTERED

//module xnorgate(in1,in2,out);
xnorgate(A0,B0,D0);
xnorgate(A1,B1,D1);
xnorgate(A2,B2,D2);
xnorgate(A3,B3,D3);

//module andgate4(in1,in2,in3,in4,out);
andgate4(D0,D1,D2,D3,K);
endmodule
```

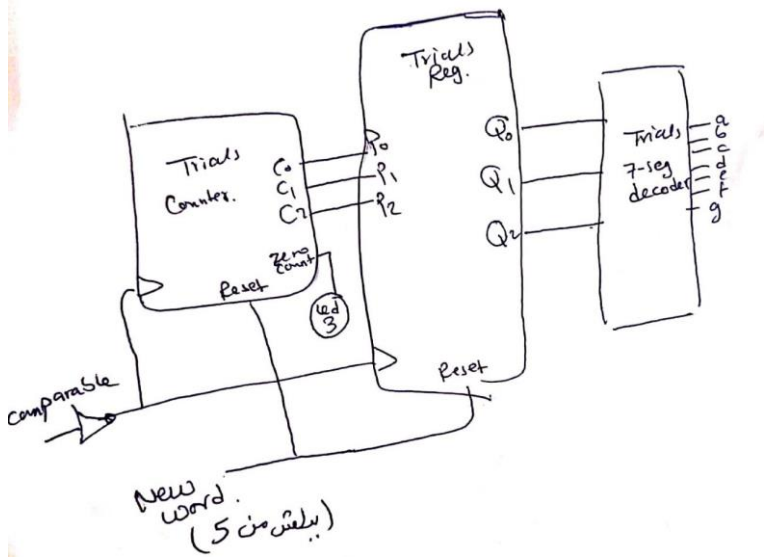
Simulation:



<u>Some helpful diagrams:</u>







PINS assignment:

Input Switches and Clock		
Start	iSW[0]	PIN_AA23
NEW_WORD	iSW[1]	PIN_AB26
ENTER	iSW[6]	PIN_AC23
Clk		PIN_E16
LETTER		
L0	iSW[8]	PIN_AD24
L1	iSW[9]	PIN_AE27
L2	iSW[10]	PIN_W5
L3	iSW[11]	PIN_V10

LEDs		
LED1 (ON/OFF)	LEDR [0]	PIN_AJ6
LED2 (WON)	LED Green[0]	PIN_W27
LED3 (LOST)	LEDR [3]	PIN_AJ4

7 segment assignment: (Letter, Trials, Guess display)

TRIALS-Display			LETTER-Display		
a4	oHEX4_D[0]	PIN_P1	a3	oHEX5_D[0]	PIN_M3
b4	oHEX4_D[1]	PIN_P2	b3	oHEX5_D[1]	PIN_L1
c4	oHEX4_D[2]	PIN_P3	c3	oHEX5_D[2]	PIN_L2
d4	oHEX4_D[3]	PIN_N2	d3	oHEX5_D[3]	PIN_L3
e4	oHEX4_D[4]	PIN_N3	e3	oHEX5_D[4]	PIN_K1
f4	oHEX4_D[5]	PIN_M1	f3	oHEX5_D[5]	PIN_K4
g4	oHEX4_D[6]	PIN_M2	g3	oHEX5_D[6]	PIN_K5
GUESS-Display					
G2-Display			G1-Display		
a2	oHEX2_D[0]	PIN_AE7	a1	oHEX1_D[0]	PIN_AG13
b2	oHEX2_D[1]	PIN_AF7	b1	oHEX1_D[1]	PIN_AE16
c2	oHEX2_D[2]	PIN_AH5	c1	oHEX1_D[2]	PIN_AF16
d2	oHEX2_D[3]	PIN_AG4	d1	oHEX1_D[3]	PIN_AG16
e2	oHEX2_D[4]	PIN_AB18	e1	oHEX1_D[4]	PIN_AE17
f2	oHEX2_D[5]	PIN_AB19	f1	oHEX1_D[5]	PIN_AF17
g2	oHEX2_D[6]	PIN_AE19	g1	oHEX1_D[6]	PIN_AD17
G0-Display					
a0	oHEX0_D[0]	PIN_AE8			
b0	oHEX0_D[1]	PIN_AF9			
c0	oHEX0_D[2]	PIN_AH9			
d0	oHEX0_D[3]	PIN_AD10			
e0	oHEX0_D[4]	PIN_AF10			
f0	oHEX0_D[5]	PIN_AD11			
g0	oHEX0_D[6]	PIN_AD12			

Category: Pin

All | Timing | Logic Options

Reserves the pin in one of seven states: as an input that is tri-stated; as an output that drives ground; as an output that drives VCC; as an output that drives an

Edit:

	To	Location	I/O Bank	I/O Standard	General Function	Special Function	Reserved
1	Start	PIN_AA23	6	3.3-V LVTTL	Row I/O	LVDS188p	
2	NEW_WORD	PIN_AB26	6	3.3-V LVTTL	Row I/O	LVDS187p	
3	ENTER	PIN_AC23	6	3.3-V LVTTL	Row I/O	LVDS194n	
4	Clk	PIN_E16	4	3.3-V LVTTL	Dedicated Clock	CLK8, LVDSCLK4n, In...	
5	L0	PIN_AD24	6	3.3-V LVTTL	Row I/O	LVDS195n/INIT_DONE	
6	L1	PIN_AE27	6	3.3-V LVTTL	Row I/O	LVDS189n	
7	L2	PIN_W5	1	3.3-V LVTTL	Row I/O		
8	L3	PIN_V10	1	3.3-V LVTTL	Row I/O	LVDS23p	
9	a4	PIN_P1	2	3.3-V LVTTL	Row I/O	LVDS36p, DPCLK0/D...	
10	b4	PIN_P2	2	3.3-V LVTTL	Row I/O	LVDS36n	
11	c4	PIN_P3	2	3.3-V LVTTL	Row I/O	LVDS37p	
12	d4	PIN_N2	2	3.3-V LVTTL	Row I/O	LVDS38p	
13	e4	PIN_N3	2	3.3-V LVTTL	Row I/O	LVDS38n	
14	f4	PIN_M1	2	3.3-V LVTTL	Row I/O	LVDS40p	
15	g4	PIN_M2	2	3.3-V LVTTL	Row I/O	LVDS40n	
16	a3	PIN_M3	2	3.3-V LVTTL	Row I/O	LVDS39p	
17	b3	PIN_L1	2	3.3-V LVTTL	Row I/O	LVDS42p	
18	c3	PIN_L2	2	3.3-V LVTTL	Row I/O	LVDS42n	
19	d3	PIN_L3	2	3.3-V LVTTL	Row I/O	LVDS45p	
20	e3	PIN_K1	2	3.3-V LVTTL	Row I/O	LVDS44p	
19	d3	PIN_L3	2	3.3-V LVTTL	Row I/O	LVDS45p	
20	e3	PIN_K1	2	3.3-V LVTTL	Row I/O	LVDS44p	
21	f3	PIN_K4	2	3.3-V LVTTL	Row I/O	LVDS49n	
22	g3	PIN_K5	2	3.3-V LVTTL	Row I/O	LVDS56n	
23	a2	PIN_AE7	8	3.3-V LVTTL	Column I/O	LVDS256p	
24	b2	PIN_AF7	8	3.3-V LVTTL	Column I/O	LVDS252n	
25	c2	PIN_AH5	8	3.3-V LVTTL	Column I/O	VREFB8N3	
26	d2	PIN_AG4	8	3.3-V LVTTL	Column I/O	LVDS255p	
27	e2	PIN_AB18	7	3.3-V LVTTL	Column I/O	LVDS213p	
28	f2	PIN_AB19	7	3.3-V LVTTL	Column I/O	LVDS213n	
29	g2	PIN_AE19	7	3.3-V LVTTL	Column I/O		
30	a1	PIN_AG13	8	3.3-V LVTTL	Column I/O	VREFB8N0	
31	b1	PIN_AE16	7	3.3-V LVTTL	Column I/O		
32	c1	PIN_AF16	7	3.3-V LVTTL	Column I/O		
33	d1	PIN_AG16	7	3.3-V LVTTL	Column I/O	VREFB7N3	
34	e1	PIN_AE17	7	3.3-V LVTTL	Column I/O		
35	f1	PIN_AF17	7	3.3-V LVTTL	Column I/O	VREFB7N2	
36	g1	PIN_AD17	7	3.3-V LVTTL	Column I/O	LVDS223p	
37	a0	PIN_AE8	8	3.3-V LVTTL	Column I/O	LVDS256n/DEV_OE	
38	b0	PIN_AF9	8	3.3-V LVTTL	Column I/O	VREFB8N2	
41	e0	PIN_AF10	8	3.3-V LVTTL	Column I/O		
42	f0	PIN_AD11	8	3.3-V LVTTL	Column I/O	LVDS244n	
43	g0	PIN_AD12	8	3.3-V LVTTL	Column I/O	LVDS240n	
44	LED1	PIN_AJ6	8	3.3-V LVTTL	Column I/O	LVDS247n	
45	LED2	PIN_W27	6	3.3-V LVTTL	Row I/O	LVDS171n	
46	LED3	PIN_AJ4	8	3.3-V LVTTL	Column I/O	LVDS251p	
47	<<new>>	<<new>>					